

0 6 . 1

A student dissected a sheep's heart. He prepared a risk assessment on:

- carrying a scalpel
- using a scalpel.

Complete **Table 3** by giving **three** control measures the student must use to reduce the risks associated with carrying **and** using a scalpel.

Table 3

Source	Hazard	Control measures to reduce risk
Sharp knife		1.
		2.
		3.

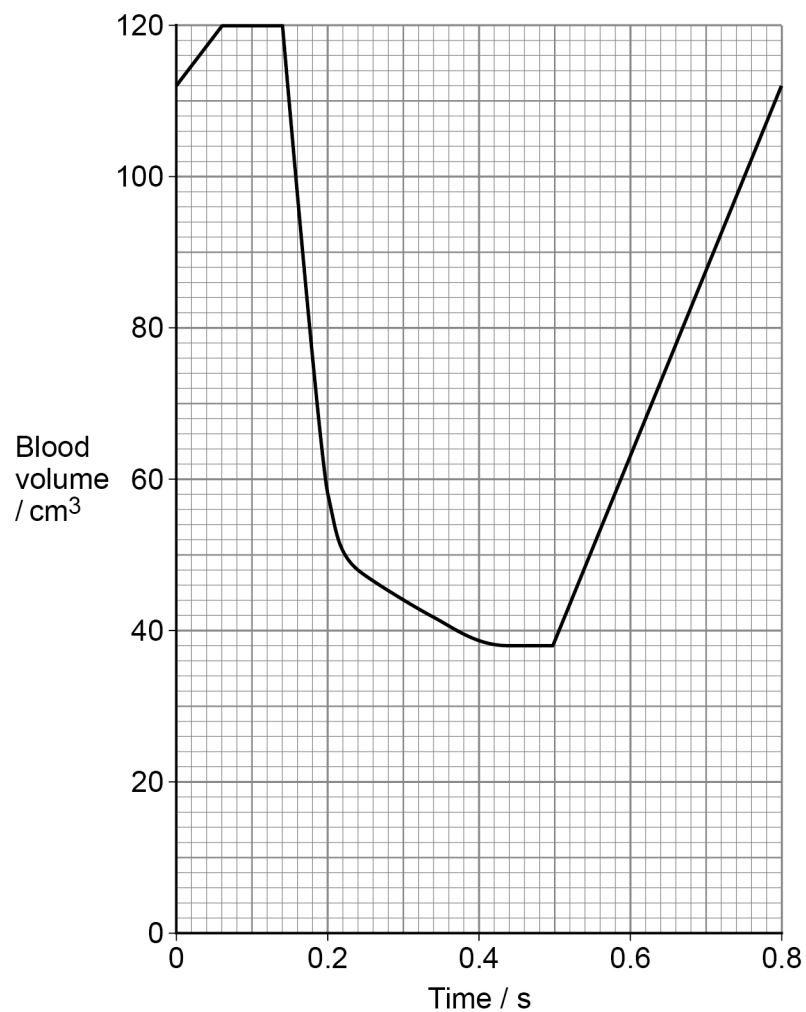
[2 marks]



0 6 . 2

A scientist measured the changes in blood volume inside the left ventricle of a mammalian heart during one heartbeat. **Figure 5** shows her results.

Figure 5



Use **Figure 5** to calculate the total volume of blood pumped from the left ventricle in 1 minute.

[2 marks]

Answer _____ cm³

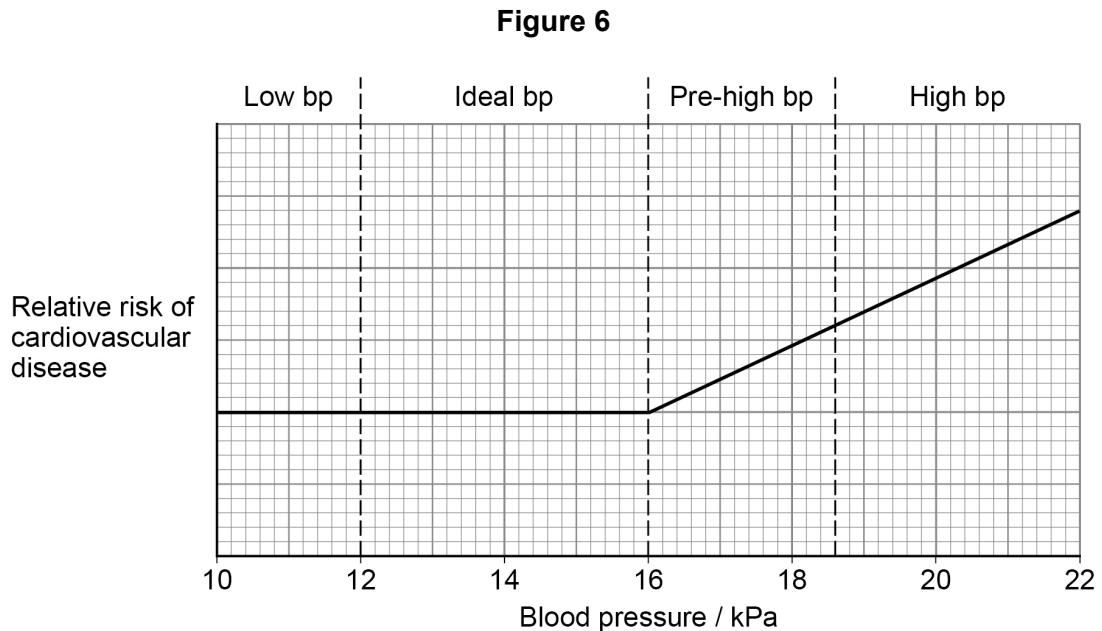
Turn over ►



Scientists investigated the correlation between patients' blood pressure (bp) and the risk of these patients developing cardiovascular disease.

They grouped the blood pressure measurements into four health categories: low, ideal, pre-high and high blood pressure.

Figure 6 shows the scientists' results.



In a separate investigation, a doctor measured the effect of blood pressure medicines on treating a large number of patients with a mean blood pressure of 22 kPa

The doctor used two different treatments.

- Treatment 1 contained one blood pressure medicine.
- Treatment 2 contained three blood pressure medicines.

He measured the change in the mean blood pressure after each treatment in these patients.

Figure 7

Change in the mean blood pressure after treatment / kPa

Treatment	Change in the mean blood pressure after treatment / kPa
Treatment 1	-1.7
Treatment 2	-4.0

Using information in **Figure 6** and **Figure 7**, evaluate the effect of these treatments on reducing the risk of developing cardiovascular disease in patients with a mean blood pressure of 22 kPa

[illegible]

8